

Server Tuning

Description

Learn about different tuning methods

- 1.Pagespeed
- 2.Gzip compression
- 3.tuned daemon
- etc

1. Apache Installation and Configuration

Description: Installed and configured Apache HTTP Server to host a sample website on the RHEL server.

Procedure

1. Installed Apache HTTP Server.

```
sudo dnf install httpd -y
```

2. Started and enabled Apache.

```
sudo systemctl enable --now httpd
```

3. Checked Apache service status.

```
sudo systemctl status httpd --no-pager
```

4. Created the website file.

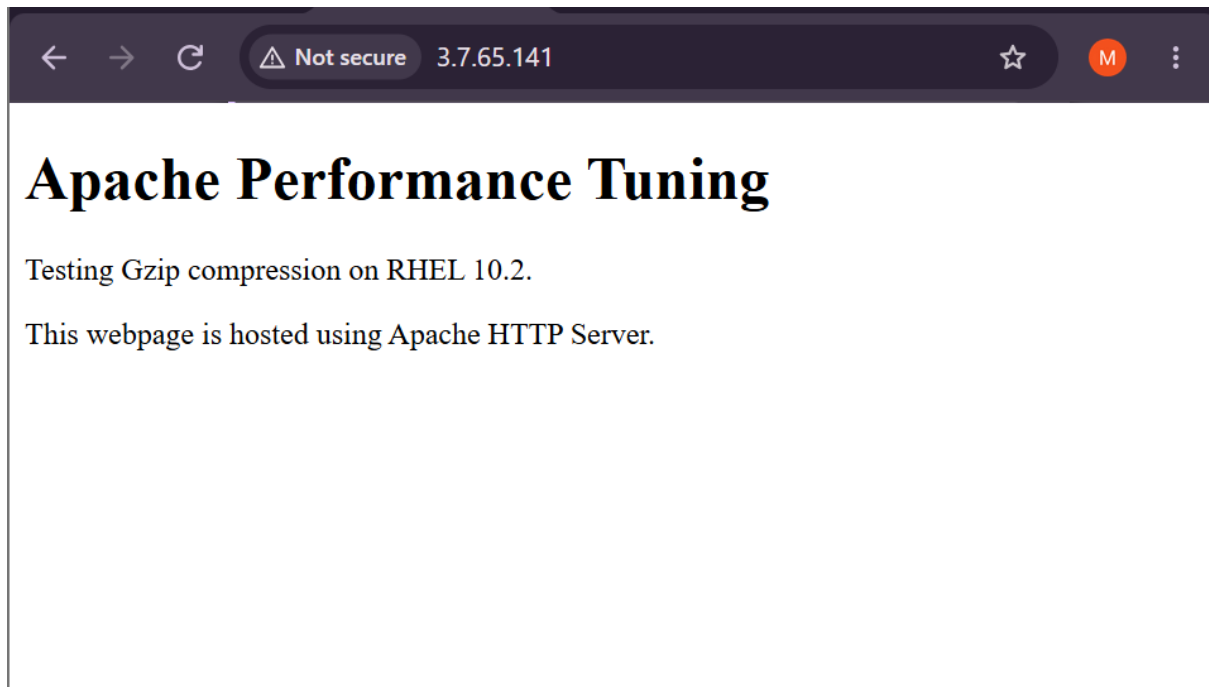
```
sudo vi /var/www/html/index.html
```

Added the sample HTML content and saved the file.

5. Set file permissions and SELinux context.

```
sudo chmod 644 /var/www/html/index.html  
sudo restorecon -v /var/www/html/index.html
```

6. Verified the website through the browser.



2. Gzip Compression

Description: Gzip compression reduces the size of website content transferred to the browser, reducing bandwidth usage and improving page loading speed.

Procedure

1. Checked the Apache Gzip module.

```
sudo httpd -M | grep -i deflate
```

Verified that deflate_module (shared) was loaded.

2. Created the Gzip configuration file.

```
sudo vi /etc/httpd/conf.d/gzip.conf
```

Added the following configuration:

```
<IfModule mod_deflate.c>
    AddOutputFilterByType DEFLATE text/html text/plain text/css
    AddOutputFilterByType DEFLATE application/javascript
    AddOutputFilterByType DEFLATE application/json
    AddOutputFilterByType DEFLATE application/xml
</IfModule>
```

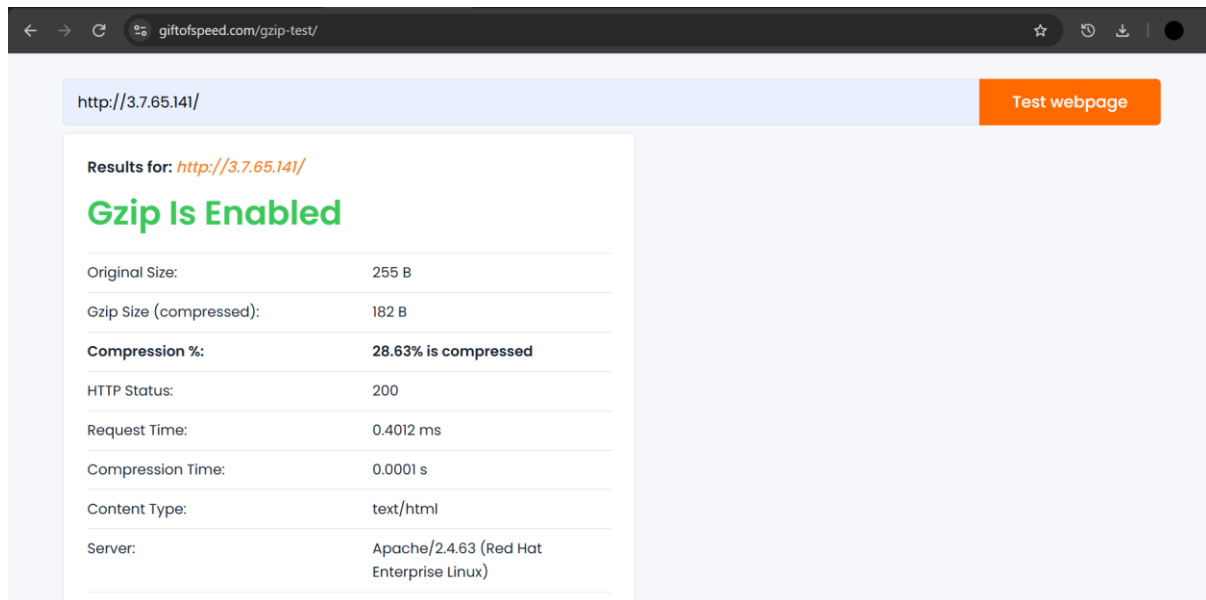
3. Tested the Apache configuration.

```
sudo apachectl configtest
```

4. Reloaded Apache.

```
sudo systemctl reload httpd
```

5. Verified Gzip compression using the website and browser-based testing.



3. Tuned Daemon

Description: Tuned is a Linux performance tuning service that applies predefined profiles to optimize system resources according to the workload.

Procedure

1. Installed the Tuned package.

```
sudo dnf install tuned -y
```

2. Started and enabled Tuned.

```
sudo systemctl enable --now tuned
```

3. Verified the Tuned service.

```
sudo systemctl status tuned --no-pager
```

4. Listed available performance profiles.

```
sudo tuned-adm list
```

5. Checked the recommended profile.

```
sudo tuned-adm recommend
```

6. Applied the recommended profile.

Used the profile name returned by the recommendation command.

Example:

```
sudo tuned-adm profile virtual-guest
```

7. Verified the active profile.

```
sudo tuned-adm active
```

```
[ec2-user@ip-172-31-15-218 ~]$ sudo systemctl status tuned --no-pager
sudo tuned-adm active
sudo tuned-adm recommend
● tuned.service - Dynamic System Tuning Daemon
   Loaded: loaded (/usr/lib/systemd/system/tuned.service; enabled; preset: enabled)
   Active: active (running) since Fri 2026-09-25 04:54:17 UTC; 8min ago
  Invocation: d67856437feb4c7398c2ca11594f9dae
     Docs: man:tuned(8)
           man:tuned.conf(5)
           man:tuned-adm(8)
 Main PID: 2147 (tuned)
    Tasks: 4 (limit: 4156)
  Memory: 15.4M (peak: 17.5M)
     CPU: 933ms
    CGroup: /system.slice/tuned.service
            └─2147 /usr/bin/python3 -Es /usr/sbin/tuned -l -P

Sep 25 04:54:16 ip-172-31-15-218.ap-south-1.compute.internal systemd[1]: Starting ...
Sep 25 04:54:17 ip-172-31-15-218.ap-south-1.compute.internal systemd[1]: Started t...
Hint: Some lines were ellipsized, use -l to show in full.
Current active profile: virtual-guest
virtual-guest
[ec2-user@ip-172-31-15-218 ~]$ |
```

```
[ec2-user@ip-172-31-15-218 ~]$ sudo tuned-adm active
Current active profile: virtual-guest
[ec2-user@ip-172-31-15-218 ~]$ sudo tuned-adm profile powersave
[ec2-user@ip-172-31-15-218 ~]$ sudo tuned-adm active
Current active profile: powersave
[ec2-user@ip-172-31-15-218 ~]$ sudo tuned-adm recommend
virtual-guest
[ec2-user@ip-172-31-15-218 ~]$ sudo tuned-adm profile virtual-guest
[ec2-user@ip-172-31-15-218 ~]$ sudo tuned-adm active
Current active profile: virtual-guest
[ec2-user@ip-172-31-15-218 ~]$ |
```

Conclusion

Successfully completed Linux server performance tuning on RHEL 10.2 using Apache HTTP Server. Installed and configured Apache, enabled Gzip compression, and applied a Tuned performance profile to optimize server performance.